AWS Cloud Security Monitoring Foundational SOC Project

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1.Project Overview

This project focuses on implementing basic cloud security monitoring in an AWS environment using native tools. The main goal was to understand how a Security Operations Center (SOC) analyst monitors and investigates cloud activities such as user logins and system events.

The lab was designed as a beginner-friendly, defensive security project using Free Tier services of Amazon Web Services. It demonstrates how security logs are generated, collected, and analyzed in a real-world cloud environment.

2.Project Objectives

The key objectives of this project are:

- Secure the AWS root account
- Create a limited-access IAM user for SOC monitoring
- Deploy a secure EC2 instance
- Enable logging using AWS CloudTrail
- Analyze authentication and activity logs

3.AWS Account Security Setup

To improve account security, Multi-Factor Authentication (MFA) was enabled for the root user. The root account was only used for administrative purposes, and no access keys were created.

This ensures that the main account remains protected and reduces the risk of unauthorized access.

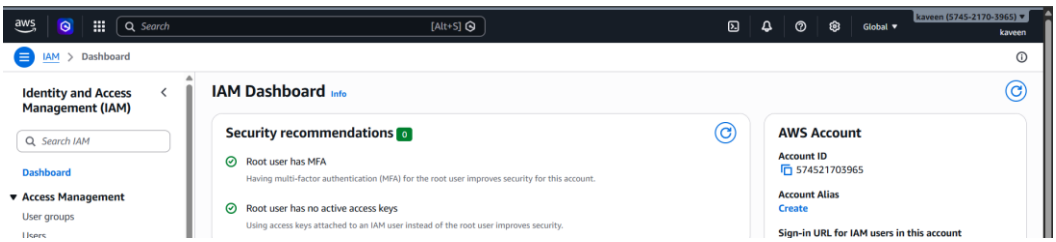


Figure 1:AWS root account security credentials page showing MFA enabled.

4.IAM Configuration and Least Privilege

An IAM user named soc-analyst was created to simulate a SOC analyst role.

The following permissions were assigned:

- SecurityAudit
- ViewOnlyAccess

These permissions allow the user to monitor logs and view configurations without making changes. This follows the principle of least privilege.

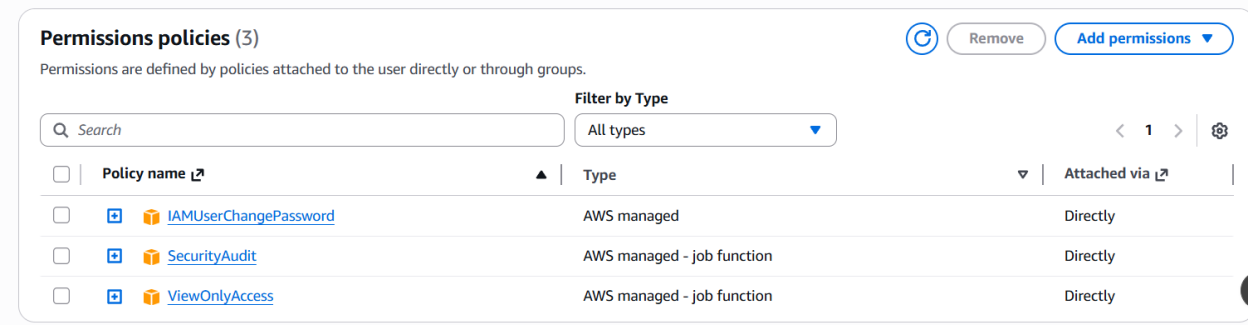


Figure 2: IAM permission Policies

Figure 3: IAM user permissions summary

5.EC2 Instance Deployment and Hardening

An EC2 instance was deployed using an Amazon Linux Free Tier eligible AMI. Key-based SSH authentication was enforced, and password-based login was not permitted. This ensures that only users with the correct private key can access the instance. The instance was launched using a privileged account, while monitoring and investigation were performed using the SOC analyst account.

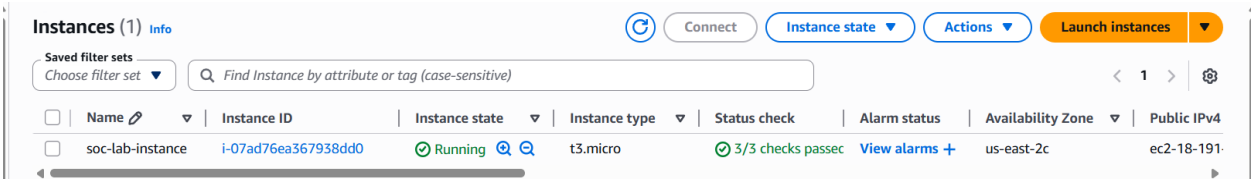


Figure 4:EC2 instance launch success page showing instance ID and running state.

6.Network Security Group Configuration

A custom security group named soc-ssh-sg was created.

Configuration:

- Allowed SSH access (port 22)
- Access restricted only to my IP address

This setup minimizes exposure and protects the instance from unauthorized access.

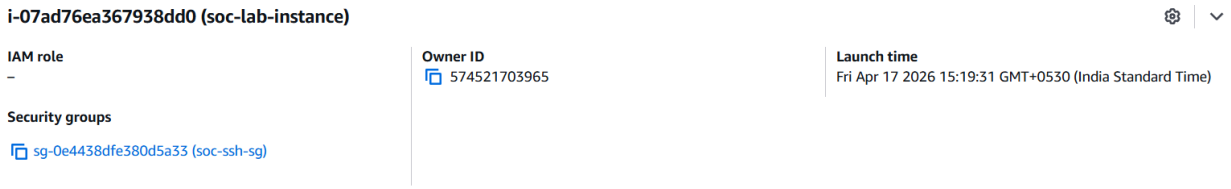


Figure 5:Security group creation page.

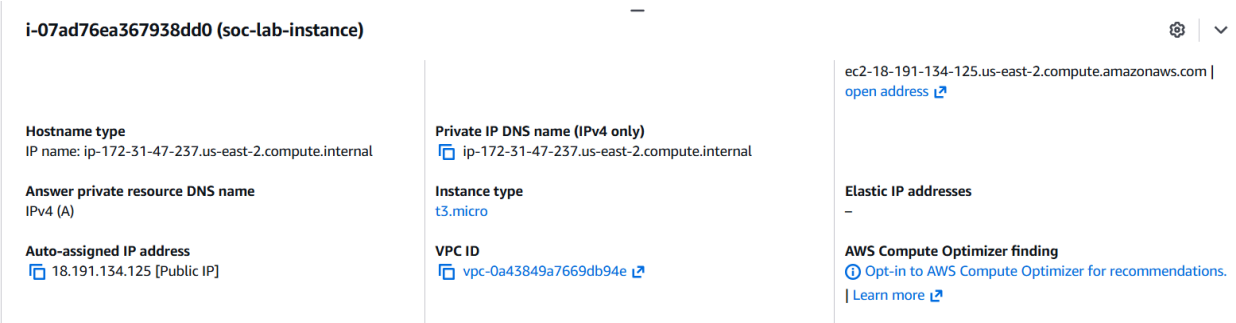


Figure 6:Security group inbound rule showing SSH access restricted to the analyst's IP

7.AWS CloudTrail Configuration

AWS CloudTrail was enabled to capture all management and authentication events.

Configuration details:

- Multi-region trail enabled
- Log file validation enabled
- Logs stored in an S3 bucket

This acts as a centralized logging system for monitoring AWS activities.

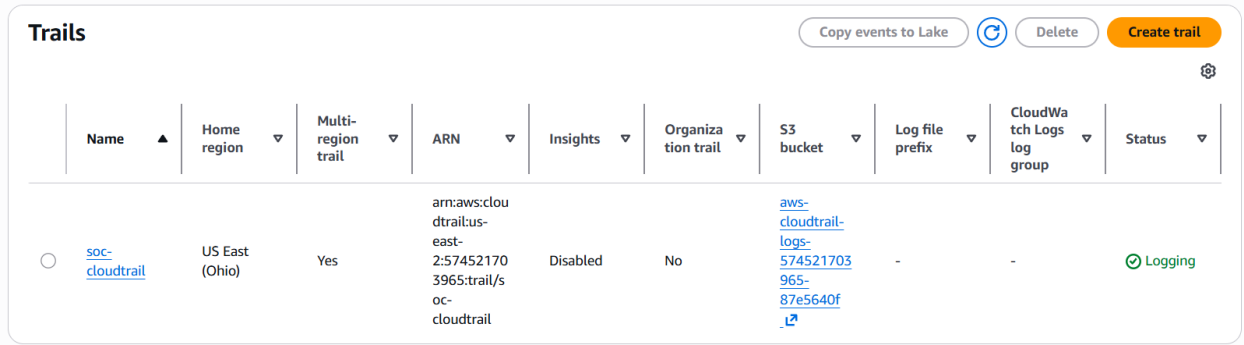


Figure 7: CloudTrail trails page showing soc-cloudtrail with logging enabled.

8.SOC Log Analysis – CloudTrail Event History

CloudTrail Event History was used to analyze system activity.

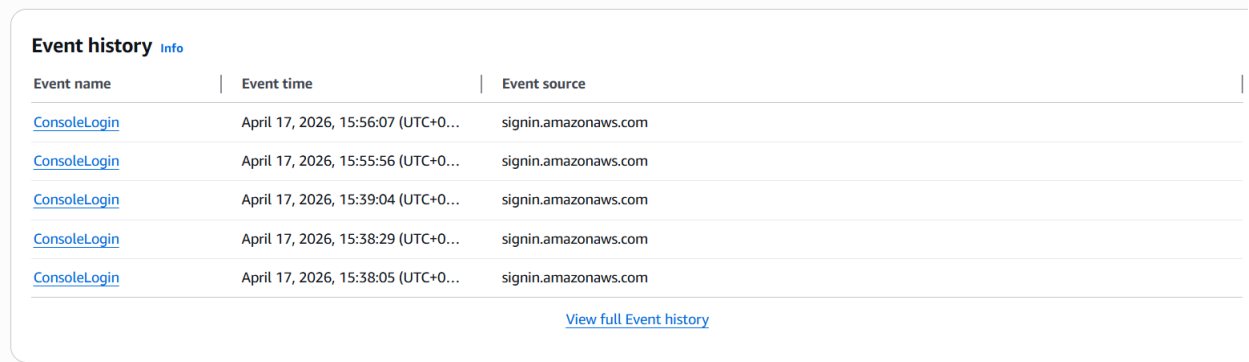
The logs were filtered using:

- Event name: ConsoleLogin

The following attributes were reviewed:

- Event time
- Username
- Source IP address
- Event source
- MFA usage

Multiple events such as login attempts and SSH key usage were successfully observed.



Event history Info

Event name	Event time	Event source
ConsoleLogin	April 17, 2026, 15:56:07 (UTC+0…	signin.amazonaws.com
ConsoleLogin	April 17, 2026, 15:55:56 (UTC+0…	signin.amazonaws.com
ConsoleLogin	April 17, 2026, 15:39:04 (UTC+0…	signin.amazonaws.com
ConsoleLogin	April 17, 2026, 15:38:29 (UTC+0…	signin.amazonaws.com
ConsoleLogin	April 17, 2026, 15:38:05 (UTC+0…	signin.amazonaws.com

View full Event history

Figure 8 :CloudTrail Event History filtered by ConsoleLogin events.

9.Detailed Event Investigation

A ConsoleLogin event from the **soc-analyst** user was selected for analysis.

Findings:

- Login was successful
- Source IP address matched expected location
- MFA was enabled
- No suspicious activity detected

Additionally, events like **SendSSHPublicKey** confirmed secure access to the EC2 instance.

The activity was classified as legitimate.

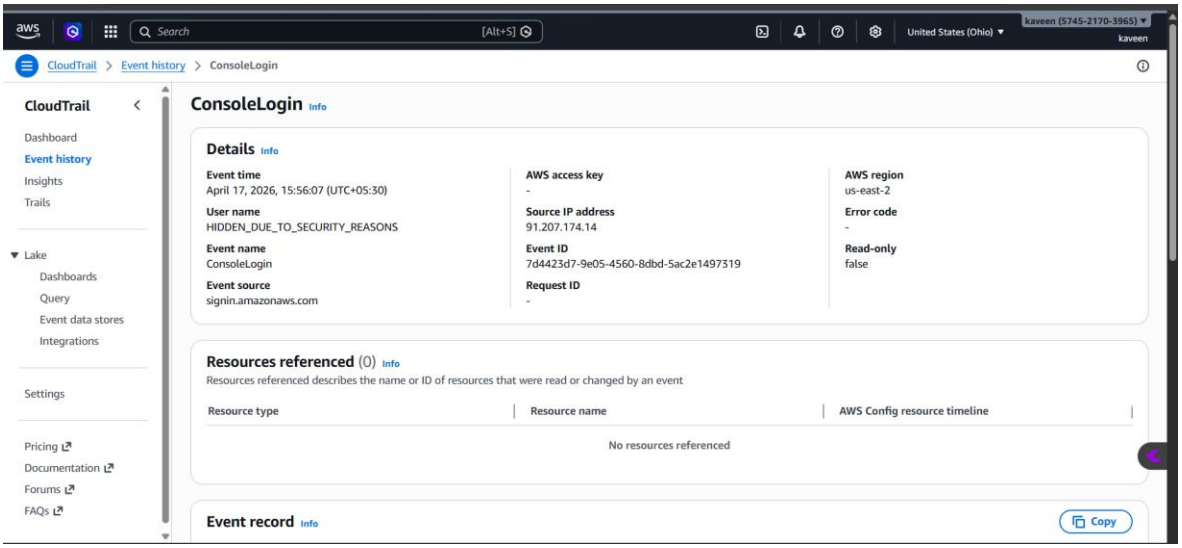


Figure 9:Detailed ConsoleLogin event showing username, source IP, and MFA enforcement

10.SOC Incident Documentation

Incident Summary:

- Event Type: AWS Console Login
- Log Source: CloudTrail
- User: soc-analyst
- Severity: Informational

Assessment: The login activity was normal, with valid credentials and MFA enabled.

Action Taken: No action required.

Status: Closed

11.Conclusion

This project provided practical experience in cloud security monitoring and log analysis. It helped in understanding how AWS services like IAM, EC2, and CloudTrail work together to provide visibility into system activities.

The lab successfully demonstrated how a SOC analyst monitors and investigates cloud-based events using native AWS tools.